

Report
Of
Machine Learning Intern

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
MCA



By

Roll Number of Student: 25SCS2040001060

Name of Student: Gokul Malav

Batch: 2026-27

Chapter 1: CANDIDATE'S DECLARATION

I, Ayush Chaudhary do hereby declare that the internship report titled “**Artificial Intelligence Intern**” has been completed by me to fulfil the requirement for the award of the degree of Master of Computer Application. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Gokul Malav

Roll No.: 25SCS2040001060

Chapter 2: ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Gokul Malav

Roll No.: 25SCS2040001060

Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	
2.	Acknowledgment	
3.	Internship Completion Certificate	
4.	Project Description 4 .1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of completion and other communication mail proof must be attached	
5.	Bibliography/References	

Chapter 3: Internship Completion Certificate



Chapter 4: Project Description

4.1 Introduction

Artificial Intelligence (AI) is an important field of computer science that focuses on developing systems capable of performing tasks that generally require human intelligence. These tasks include learning from data, recognizing patterns, understanding language, analyzing information, and making decisions.

The one-month Artificial Intelligence Internship was designed to provide foundational knowledge and practical exposure to various AI concepts, tools, and techniques. The internship covered Python programming for AI development, data preprocessing and visualization, machine learning, deep learning, Convolutional Neural Networks (CNNs), Natural Language Processing (NLP), and Reinforcement Learning.

The internship followed a structured, week-wise learning approach, progressing from fundamental AI concepts to practical implementation of different AI techniques.

During the first week, the focus was on understanding AI concepts, Python programming for AI development, and preparing and visualizing data. The second week introduced supervised machine learning, model evaluation, and optimization techniques.

The third week focused on deep learning and neural networks, particularly CNNs for image data. It included understanding image recognition applications and building and training a basic CNN model for image classification.

The fourth week covered advanced AI applications, including Natural Language Processing and Reinforcement Learning. Practical learning included text preprocessing, sentiment analysis, implementing basic NLP tasks using NLTK and spaCy, and developing a simple Q-learning model. Model performance evaluation was also covered using metrics such as accuracy, precision, recall, and F1-score.

Overall, the internship provided an opportunity to understand the fundamental principles of AI and explore how different AI techniques can be applied to solve computational problems.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is an ISO 9001:2015 certified IT services, consultancy, and technical skill development enterprise. The company delivers enterprise-grade hosting solutions, cloud integrations, and custom software development across global markets while maintaining an active corporate training ecosystem recognized by national and international quality bodies, including the All-India Council for Technical Education (AICTE) and the International Certification Accreditation Council (ICAC).

Operating as both a tech-solutions provider and a structured skill-development platform, Codec Technologies bridges the gap between traditional academic curricula and modern industry expectations. The organization provides interns with exposure to live datasets, production-grade tools, and structured project frameworks across emerging tech domains such as Data Analytics, Machine Learning, Web Engineering, and Cloud Infrastructure.

The corporate culture at Codec Technologies emphasizes hands-on software development, rigorous documentation standards, continuous mentorship, and agile project execution. The firm's internal structure is broadly partitioned into four operational units:

- **Enterprise IT Consultancy & Digital Solutions:** Responsible for engineering scalable software systems, web applications, and automated workflow solutions for client organizations across varied business sectors.
- **Data Analytics & Business Intelligence Division:** Specializes in processing large volumes of operational data, designing relational and dimensional data models, and constructing interactive BI dashboards that empower enterprise leaders to make data-driven decisions.
- **Cloud Infrastructure & Managed Services:** Focuses on cloud migration strategies, high-performance hosting environments, database administration, and IT security protocols.
- **Corporate Training & Skill Acceleration Wing:** Manages AICTE-recognized internship programs, technical bootcamps, and industrial project-based modules designed to train emerging developers in enterprise-level tech stacks.

During the internship period, the Data Analytics & Business Intelligence Division served as the primary reporting unit. Working under the guidance of senior technical leads, interns executed end-to-end data pipelines—ranging from initial programmatic data manipulation in Python to executive dashboard design in Microsoft Power BI—ensuring adherence to modern data architecture and corporate analytics standards.

4.3 Problem Statement

Artificial Intelligence is increasingly being applied to problems involving data analysis, image recognition, text understanding, and intelligent decision-making. However, developing AI-based solutions requires an understanding of different algorithms, data preparation techniques, model training procedures, and performance evaluation methods.

Beginners in Artificial Intelligence may face challenges in understanding how these concepts are connected and how they can be implemented using programming languages and relevant libraries.

Some of the key challenges include:

- **Understanding AI Fundamentals:** Developing a clear understanding of Artificial Intelligence concepts and their practical applications.
- **Data Preparation:** Preparing datasets through preprocessing and visualization before applying machine learning and deep learning algorithms.
- **Model Development:** Understanding supervised learning techniques and the process of training and optimizing machine learning models.
- **Image Recognition:** Understanding how neural networks and CNNs can be used to process image data and perform image classification.
- **Text Analysis:** Learning how NLP techniques can be used to preprocess text and perform tasks such as sentiment analysis.
- **Intelligent Decision-Making:** Understanding how Reinforcement Learning algorithms, such as Q-learning, can be used to learn actions through interaction with an environment.
- **Model Evaluation:** Understanding performance metrics such as accuracy, precision, recall, and F1-score to evaluate classification models.

Therefore, the primary problem addressed through this internship was the need for structured foundational and practical learning in different areas of Artificial Intelligence, enabling the understanding of AI workflows and their applications in real-world problem-solving.

4.4 Project Objectives

The main objective of the internship was to gain practical knowledge of Artificial Intelligence and understand how AI techniques can be applied to real-world problems. The training focused on learning fundamental AI concepts, working with data, developing machine learning models, and understanding the basics of deep learning and Natural Language Processing.

The specific objectives were:

- To understand the fundamental concepts of Artificial Intelligence and its applications in different domains.
- To learn Python programming concepts relevant to AI development.
- To understand data preprocessing and visualization techniques for preparing datasets.
- To gain knowledge of supervised machine learning algorithms and understand model evaluation and optimization.
- To understand the fundamentals of neural networks and deep learning.
- To explore the architecture and applications of Convolutional Neural Networks for image recognition.
- To build and train a basic CNN model for image classification.
- To understand the fundamentals of Natural Language Processing and its applications in text analysis.
- To learn text preprocessing and sentiment analysis techniques using Python libraries such as NLTK and spaCy.
- To understand Reinforcement Learning concepts and implement a simple Q-learning model.
- To evaluate model performance using accuracy, precision, recall, and F1-score.
- To develop a foundation for further learning and practical development in Artificial Intelligence.

4.5 Scope of the Project

The scope of the internship was to provide foundational knowledge and practical exposure to various areas of Artificial Intelligence. The training covered the complete learning process, beginning with basic AI concepts and Python programming and progressing toward machine learning, deep learning, NLP, and Reinforcement Learning.

During the internship, practical training included the following two projects:

1. Graduate Admission Prediction using ANN

This training project focused on understanding how Artificial Neural Networks can be used for prediction tasks. It provided practical exposure to the application of ANN techniques to a graduate admission prediction problem and helped develop an understanding of how input data can be used to generate predictions through a trained model.

2. Handwritten Digit Classification using ANN

This training project focused on the classification of handwritten digits using Artificial Neural Networks. It provided practical exposure to using neural networks for image-related classification tasks and helped in understanding how a model can learn patterns from input data to identify digit classes.

These projects were completed as part of the internship training. They were not independent or personal projects assigned separately from the training program.

The internship also introduced CNNs for image data, NLP techniques for text processing, and Q-learning as a basic Reinforcement Learning approach.

4.6 Technologies and Tools Used

The internship provided exposure to the following technologies and tools:

Technology / Tool	Purpose
Python	Programming language used for AI development and practical exercises.
Data Preprocessing	Preparing and cleaning data for analysis and model training.
Data Visualization	Understanding and presenting patterns in data.
Artificial Neural Networks (ANN)	Learning and implementing prediction and classification techniques.
Convolutional Neural Networks (CNN)	Understanding deep learning techniques for image data.
NLTK	Learning text preprocessing and basic NLP tasks.
spaCy	Exploring NLP tools and text-processing techniques.
Reinforcement Learning	Understanding the fundamentals of learning through rewards.
Q-learning	Implementing a basic Reinforcement Learning approach.
Model Evaluation Metrics	Evaluating models using accuracy, precision, recall, and F1-score.

4.7 System Architecture

The internship training involved learning a general AI model development workflow.

The following architecture represents the basic workflow followed in the training-based prediction and classification tasks.



Architecture Description

1. **Input Data:** The data required for the prediction or classification task is provided to the system.
2. **Data Preprocessing:** The input data is prepared for model development through preprocessing.
3. **Data Preparation:** The processed data is organized for training and evaluation.
4. **ANN Model:** An Artificial Neural Network is used to learn patterns from the prepared data.
5. **Model Training:** The model learns from the training data.
6. **Model Evaluation:** The trained model is evaluated using suitable performance metrics.
7. **Output:** The model generates a prediction or classification result.

4.8 Methodology

The internship followed a week-wise learning methodology, combining theoretical concepts with practical exercises and training-based projects.

Week 1: Introduction to AI and Python

The first week focused on the fundamentals of Artificial Intelligence and its applications. Python programming concepts relevant to AI development were introduced. The training also covered data preprocessing and visualization to understand how data is prepared and explored before applying AI techniques.

Week 2: Supervised Learning

The second week focused on supervised learning concepts and their applications. The training introduced model evaluation and optimization techniques to understand how machine learning models are assessed and improved.

Week 3: Deep Learning and ANN-Based Projects

The third week introduced the basics of deep learning and neural networks. The training included CNN concepts and their applications in image recognition, along with building and training a basic CNN model for image classification.

The following ANN-based projects were also included in the practical training:

Graduate Admission Prediction using ANN: Practical exposure to applying ANN techniques to a graduate admission prediction task.

Handwritten Digit Classification using ANN: Practical exposure to using ANN for handwritten digit classification.

These projects helped connect neural network concepts with practical prediction and classification tasks.

Week 4: NLP and Reinforcement Learning

The fourth week covered the fundamentals of Natural Language Processing, including text preprocessing and sentiment analysis. Simple NLP tasks were introduced using NLTK and spaCy.

The training also introduced Reinforcement Learning and included the implementation of a basic Q-learning model. Model performance evaluation was covered using accuracy, precision, recall, and F1-score.

4.9 Expected Outcomes

The internship helped build foundational knowledge of Artificial Intelligence and provided practical exposure to different AI techniques.

The major learning outcomes were:

Developed an understanding of AI concepts and their applications.

Gained exposure to Python programming for AI development.

Learned the importance of data preprocessing and visualization.

Understood the fundamentals of supervised learning and model evaluation.

Gained practical exposure to Artificial Neural Networks through the two training-based projects.

Understood the basic concepts of deep learning and CNNs for image classification.

Learned the fundamentals of NLP and text preprocessing using NLTK and spaCy.

Gained introductory exposure to Reinforcement Learning and Q-learning.

Understood the use of accuracy, precision, recall, and F1-score for evaluating model performance.

Overall, the internship provided an opportunity to connect theoretical AI concepts with practical learning activities and develop a foundation for further study in Artificial Intelligence and Machine Learning.

4.10 Internship Proof



INTERNSHIP OFFER LETTER

Dear Gokul Malav,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 08/07/2026 to 07/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

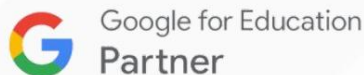
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 08/07/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



www.codectechnologies.in



support@codectechnologies.in



Chandivali IT hub, Mumbai

Chapter 5: Bibliography and References

Academic Literature & Books

1. Russell, S. and Norvig, P., 2021. Artificial Intelligence: A Modern Approach. 4th ed. Pearson.
2. Géron, A., 2022. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. 3rd ed. O'Reilly Media.
3. Goodfellow, I., Bengio, Y. and Courville, A., 2016. Deep Learning. MIT Press.
4. Bird, S., Klein, E. and Loper, E., 2009. Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. O'Reilly Media.
5. Sutton, R.S. and Barto, A.G., 2018. Reinforcement Learning: An Introduction. 2nd ed. MIT Press.

Journal Articles & Research Papers

6. LeCun, Y., Bottou, L., Bengio, Y. and Haffner, P., 1998. Gradient-based learning applied to document recognition. Proceedings of the IEEE, 86(11), pp.2278–2324.
7. Mnih, V. et al., 2015. Human-level control through deep reinforcement learning. Nature, 518, pp.529–533.
8. Hochreiter, S. and Schmidhuber, J., 1997. Long short-term memory. Neural Computation, 9(8), pp.1735–1780.

Industry Standards & Corporate Manuals

9. Python Software Foundation. Python Documentation. Available at: <https://docs.python.org/3/>
10. Scikit-learn Developers. Scikit-learn User Guide. Available at: https://scikit-learn.org/stable/user_guide.html
11. TensorFlow. TensorFlow Documentation. Available at: <https://www.tensorflow.org/learn>
12. NLTK Project. Natural Language Toolkit Documentation. Available at: <https://www.nltk.org/>
13. Explosion AI. spaCy Documentation. Available at: <https://spacy.io/usage>
14. NumPy Developers. NumPy Documentation. Available at: <https://numpy.org/doc/>